

Operations Research 2026, Final Project

Optimization-Based Meal Recommendation for Convenience Stores

A MILP and GRASP-LS Framework for Personalized Meal Recommendation

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Background & Motivation

- ▶ Convenience stores are the most frequent real-time meal channel for students and office workers.
- ▶ One pick juggles many goals at once: **cost, calories, protein, sugar, sodium, satiety**.
- ▶ Bundle promotions (e.g. rice ball + drink) change the **real** cost — the cheapest combo is not obvious.
- ▶ Ranking by a single metric is not enough $\Rightarrow$ we need **constrained optimization**.

Decision Maker: Inputs, Outputs, Fallback

Decision maker: the user.

Input

- ▶ Scenario: light / regular / late-night · Goal: maintain / cut / bulk
- ▶ Profile: sex, age, height, weight, activity · Budget cap B

Output

- ▶ Recommended single items + promo bundles, with expected nutrition and total cost

Fallback

- ▶ If infeasible: the closest alternative via soft-constraint relaxation

Preprocessing: Turning User Profile into Meal Limits

We convert each user profile into personalized calorie, protein, and item-count limits for a single meal.

Profile $\rightarrow$ BMR $\rightarrow$ TDEE $\rightarrow$ goal-adjusted calories $\rightarrow$ single-meal limits

$$\text{BMR} = \begin{cases} 10W + 6.25H - 5A + 5, & \text{male} \\ 10W + 6.25H - 5A - 161, & \text{female} \end{cases} \Rightarrow \text{TDEE} = \text{BMR} \cdot AF$$

$$\text{Target}_{\text{Cal}} = \text{TDEE} + \delta_{\text{goal}}, \quad \delta \in \{0, -500, +300\} \text{ (maintain/cut/bulk)}$$

Scenario-based meal scaling

Scenario	$\rho^{\min}$	$\rho^{\max}$	ρ (protein)	$N^{\max}$
Light	0.15	0.25	0.20	3
Regular	0.30	0.45	0.40	5
Late-night	0.05	0.15	0.10	3

Binary variables

$$a_i \in \{0, 1\} \text{ (single item } i), \quad y_k \in \{0, 1\} \text{ (bundle } k)$$

- ▶ Auxiliary variable x_i prevents the same item from being selected twice: $x_i = a_i + \sum_{k: i \in S_k} y_k \leq 1$
- ▶ Non-negative slacks $s_P, s_F^\pm, s_C^\pm, s_{\text{Sug}}, s_{\text{Sod}} \geq 0$ relax the soft targets

Tiered Memory: Penalizing Repetition (MLFQ-style)

Inspired by MLFQ, recently recommended items receive stronger penalties, and older items gradually fade.

- ▶ Tier weights $\omega = (5, 2.5, 1, 0.3)$ decay with recency; a re-picked item jumps back to the top tier
- ▶ h_i = **same-item** penalty; g_i = **same-category** penalty (e.g. all soy milks)

$$h_i = \sum_r \omega_{t(r)} \mathbf{1}(i \in \text{rec}_r), \quad g_i = \sum_r \omega_{t(r)} \mathbf{1}(\text{role}(\text{rec}_r) = \text{role}(i))$$

MILP — Objective & Constraints

Objective (minimize):

$$\alpha \underbrace{\left(\sum c_i a_i + \sum c_k^{\text{promo}} y_k \right)}_{\text{total cost}} - \beta \sum \text{pro}_i x_i + \gamma \sum h_i x_i + \gamma_{\text{cat}} \sum g_i x_i + \sum_* \lambda_* s_*$$

- ▶ **Hard constraints:** budget, calorie range, mutual exclusion, and item count
- ▶ **Meal-structure constraints:** drink/dessert caps and regular-meal composition
- ▶ **Safety constraint:** sodium $\leq 1.5 \times$ recommended cap
- ▶ **Soft constraints (slack):** protein floor, fat/carb ranges, sugar/sodium caps

Weights control the relative importance of cost, protein, repetition, and soft-constraint violations.

Why a Heuristic in Addition to Gurobi?

- ▶ The project requires our **own algorithm**, benchmarked against the exact optimum
- ▶ **License-free** and edge-deployable; extensible to multiple store chains
- ▶ **Anytime**: can stop early and still return the best meal found — suitable for interactive deployment with controllable time limits

- ▶ **Greedy Randomized Construction:** build an RCL from composite score σ and sample; structure-aware (main first, drink/dessert caps)
- ▶ **Local Search:** first-improvement over swap / drop / add / bundle-swap
- ▶ **Multi-start:** 25 random seeds $\rightarrow$ a solution pool
- ▶ **Deployment:** sample the ϵ -best set ($\epsilon = 8$); fall back to MILP if a tight budget leaves GRASP-LS infeasible

Composite Score: How the Heuristic Ranks Candidates

$$\sigma(p) = \beta \cdot pro_p - 0.06\alpha \cdot c_p - \gamma h_p - 0.6 \frac{cal_p}{Cal_m^{\max}} - \lambda_{Sod} \frac{sod_p}{Sodium_m^{\max}} - \lambda_{Sug} \frac{sug_p}{Sugar_m^{\max}} + B_{cat}(p)$$

- ▶ Rewards **protein** and **structural fit** (mains, protein sources)
- ▶ Penalizes **cost, excess calories, sodium, sugar, and repetition**
- ▶ Treats **bundles as super-items**, prioritized by discount

Dataset: Real + Random

Real-world data (live-scraped)

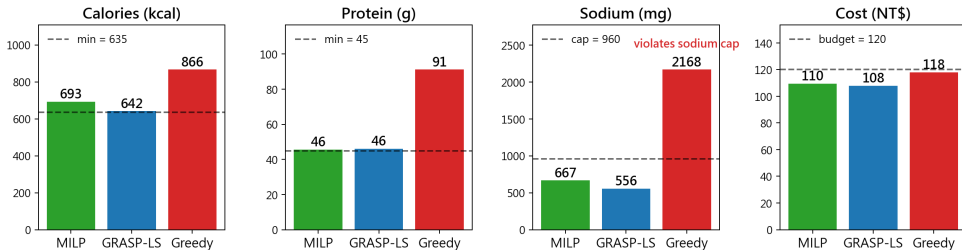
- ▶ FamilyMart food-safety API (nutrition) + promo page (tiered prices)
- ▶ 581 items $\rightarrow$ **488** after filtering family-size packs
- ▶ 124 fresh-food at real tiered prices; rest **market-calibrated**
- ▶ 6 users $\times$ 3 scenarios = **18 real instances**

Random instances

- ▶ $|I|$ up to 300, $|K|$ up to 90
- ▶ 5 seeds per size = **30 instances**
- ▶ Nutrition & discounts sampled in realistic ranges

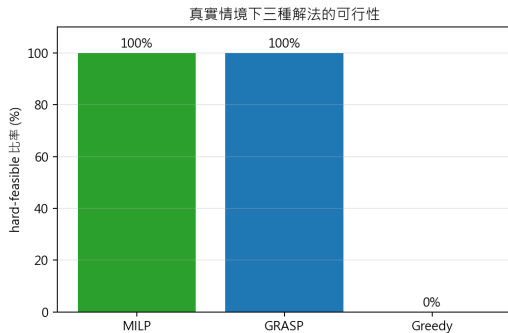
Real Scenario: MILP vs. GRASP-LS vs. Greedy

U1 (male · cut · regular): nutrition & cost across solvers



- MILP and GRASP-LS both return a **structurally complete, feasible** meal (NT\$110, all targets met)
- Greedy skips a main and breaks the sodium cap ⇒ **infeasible**

18 Real Instances — Feasibility & Optimality Gap



- ▶ **MILP: global optimum** in all cases
- ▶ **GRASP-LS: 100% hard-feasible**; average optimality gap 7.1%
- ▶ 12/18 instances match the MILP optimum exactly
- ▶ **Greedy: infeasible** under structure + sodium constraints
- ▶ Non-zero gaps appear mainly in regular-meal scenarios

Interactive Web Application (Flask)



便利商店最佳化餐食推薦

114-2 Operations Research Final Project · Group 13



1 輸入你的條件

性別	年齡
男	22
身高 (cm)	體重 (kg)
175	70
活動量	健康目標
中 (每週運動 1-3 次)	維持現狀
情境模式	預算上限 (NT\$)
正餐	120

✓ 進階：模型權重設定

成本權重	蛋白質權重
α	β
1	0.5
多樣性權重	求解器
γ	GRASP-LS (本研究啟發式, ✓)
6	

推薦這一餐

近期紀錄

2 推薦結果

本餐花費 NT\$115	求解器 GRASP	目標值 179.64	計算耗時 1596.9 ms
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推薦清單

香蔥經典嫩雞胸

\$45

蛋白

154	22.4	563
熱量 kcal	蛋白 g	鈉 mg

經典黑炫牛奶酪燻沙

\$50

蛋白

472	7.1	174
熱量 kcal	蛋白 g	鈉 mg

讚岐烏龍麵

\$20

主食

162	3.8	216
熱量 kcal	蛋白 g	鈉 mg

單餐限制達成度

熱量 (kcal) 達標
788.0 / 785-1178

蛋白質 (g) 達標
33.3 / 31-62

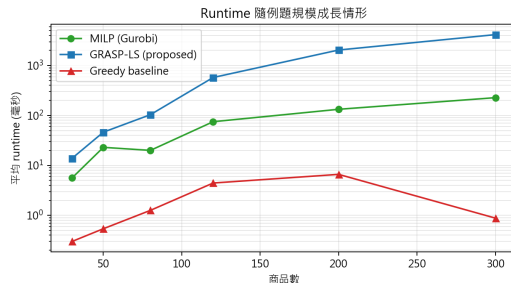
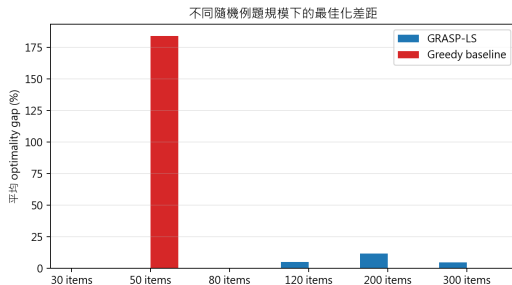
脂肪 (g) 達標
31.6 / 23-35

碳水 (g) 偏低
92.6 / 118-170

糖 (g) 偏高
40.2 / 0-26

鈉 (mg) 達標
953.0 / 0-960

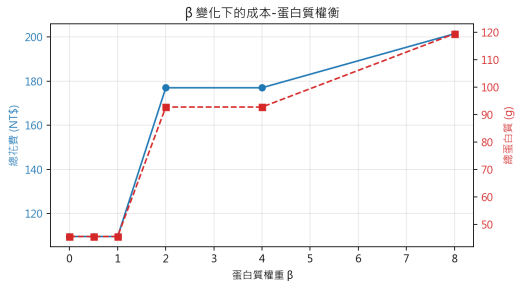
Random Instances: Scalability & Gap



- ▶ **GRASP-LS** is robust: feasible in **28/30** instances with a 4.9% average optimality gap; even under tight-budget $\times$ high-protein stress, the gap stays $\leq 7.4\%$
- ▶ **Greedy** mostly infeasible under realistic constraints
- ▶ GRASP-LS is slower here, but remains **license-free**, **interruptible**, and easier to adapt to larger deployment settings

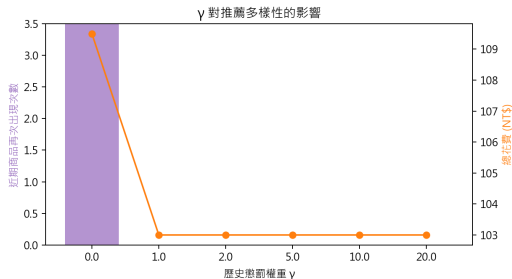
Sensitivity Analysis: Weights β and γ

β : cost-protein trade-off



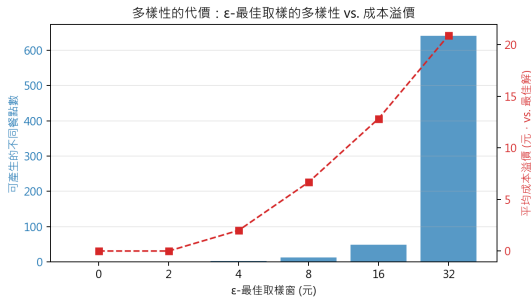
Higher $\beta \rightarrow$ more protein at higher cost
(45.6g $\rightarrow$ 119.5g).

γ : diversity



$\gamma \geq 1$ removes repetition without increasing cost; here, diversity even lowers cost from NT\$109.5 to NT\$103.

The Price of Diversity



$$\mathcal{S}_\epsilon = \{s : z(s) \leq z^* + \epsilon\}$$

- ▶ Near-optimal diversity is **cheap**
- ▶ $\epsilon = 8 \rightarrow$ **12** on-target meals, only +NT\$6.7 on average
- ▶ This lets the app **rotate diverse meals** while staying very close to the optimum

Conclusions — Academic Contributions

- ▶ **MILP formulation** with promo bundles, personalized nutrition, meal structure, and a diversity penalty
- ▶ **GRASP-LS**: high feasibility with small optimality gaps (**7.1%** / **4.9%**), anytime
- ▶ **Real-data validation**: 488 items + a working web prototype

Conclusions — Practical Insights

- ▶ Simple greedy ranking **fails** under realistic constraints (**0% feasible**)
- ▶ **Role tagging** makes recommendations meal-like, not nutrient piles
- ▶ **Real value**: compared with a simulated blind pick, all-target attainment increases from $\sim 4\%$ to 100% at comparable cost
- ▶ **Personalization is practical**: β controls the cost–protein preference, while diversity can be added at low cost

Future Work

- ▶ **Multi-day planning** with restock / inventory dynamics
- ▶ **OCR-based** in-store product updates
- ▶ **Online learning** from user click / reject feedback
- ▶ **Multi-chain integration** (7-11 / Hi-Life / FamilyMart)
- ▶ **Stronger neighborhoods** (2-2 swap, ejection chain) for GRASP-LS

Thank you!